Artwork Document Information

Product : T-GLUCOXIT XR 750MG TAB 6X10'S [FRONT]
SAP No. :
Date : 09 February 2026

Artwork Type [Please Tick /]

- [ ] Unit Box
- [x] Leaflet
- [ ] Aluminium Foil
- [ ] Generic Box
- [ ] Outer Box
- [ ] Fix-A-Form
- [ ] Sachet
- [ ] Label
- [ ] Shipper Carton
- [ ] Others : _______________

Sign / Stamp

Document Category [Please Tick /]

- [x] RA/NPRA Artwork
- [ ] Recirculation Artwork
- [ ] Approval Copy Artwork
- [ ] Clean Copy
- [ ] Internal Artwork Circulation
- [ ] Previous Approved Artwork
- [ ] Draft Artwork
- [ ] Others : ______________________________

Remarks

1. Font : Times New Roman (Spacing - 11.3pt / Size - 8pt)
2. Dimension : 297mm(w) x 420mm(h)

Colour

Black

Colours shown on this proof are to indicate correct colour separations.

Pharmacode / QR Code / Others Info

Z2

X2 W2 Y2

Label	Description	Standard
W2	Width of thin code bar	0.35 mm
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297 mm

420 mm

GLUCOXIT XR TABLET

PRODUCT DESCRIPTION
Glucoxit XR Tablet 750mg:
Capsule shape, white to off white, biconvex, with '750' marking on one side and plain on the other side, uncoated tablet.

QUALITATIVE AND QUANTITATIVE COMPOSITION
Each extended-release tablet contains:
Glucoxit XR Tablet 750mg:
Metformin hydrochloride 750mg

INDICATIONS
Reduction in the risk or delay of the onset of type 2 diabetes mellitus in adult, overweight patients with Impaired Glucose Tolerance and/or Impaired Fasting Glucose, and/or increased HbA1C who are:
- at high risk for developing overt type 2 diabetes mellitus and
- still progressing towards type 2 diabetes mellitus despite implementation of intensive lifestyle change for 3 to 6 months

Treatment with Glucoxit XR must be based on a risk score incorporating appropriate measures of glycaemic control and including evidence of high cardiovascular risk.

Lifestyle modifications should be continued when metformin is initiated, unless the patient is unable to do so because of medical reasons.

Treatment of type 2 diabetes mellitus in adults, when dietary management and exercise alone does not result in adequate glycaemic control. Glucoxit XR may be used as monotherapy or in combination with other oral antidiabetic agents, or with insulin. A reduction of diabetic complications has been shown in overweight type 2 diabetic patients treated with immediate release metformin as first-line therapy after diet failure.

PHARMACOLOGICAL PROPERTIES
Pharmacodynamic properties
ORAL ANTI-DIABETIC
(A10BA02: Gastrointestinal tract and metabolism)

Metformin is a biguanide with anti-hyperglycaemic effects, lowering both basal and postprandial plasma glucose. It does not stimulate insulin secretion and therefore does not produce hypoglycaemia.

Metformin may act via 3 mechanisms:
1. Reduction of hepatic glucose production by inhibiting gluconeogenesis and glycogenolysis.
2. In muscle by increasing insulin sensitivity, improving peripheral glucose uptake and utilisation.
3. And delay of intestinal glucose absorption

Metformin stimulates intracellular glycogen synthesis by acting on glycogen synthase.

Metformin increases the transport capacity of all types of membrane glucose transporters (GLUT).

Pharmacokinetic properties
Absorption
After an oral dose of the extended-release tablet, metformin absorption is significantly delayed compared to the immediate release tablet with a T_{max} at 7 hours (T_{max} for the immediate release tablet is 2.5 hours). At steady state, similar to the immediate release formulation, C_{max} and AUC are not proportionally increased to the administered dose. The AUC after a single oral administration of 2000mg of metformin extended-release tablets is similar to that observed after administration of 1000mg of metformin immediate release tablets.

Intrasubject variability of C_{max} and AUC of metformin extended release is comparable to that observed with metformin immediate release tablets. When the extended-release tablet is administered in fasting conditions the AUC is decreased by 30% (both C_{max} and T_{max} are unaffected).

Metformin absorption from the extended-release formulation is not altered by meal composition. No accumulation is observed after repeated administration of up 2000 mg of metformin as extended-release tablets.

Distribution
Plasma protein binding is negligible. Metformin partitions into erythrocytes. The blood peak is lower than the plasma peak and appears at approximately the same time. The red blood cells most likely represent a secondary compartment of distribution. The mean V_d ranged between 63 – 276 L.

Metabolism
Metformin is excreted unchanged in the urine. No metabolites have been identified in humans.

Elimination
Renal clearance of metformin is > 400ml/min, indicating that metformin is eliminated by glomerular filtration and tubular secretion. Following an oral dose, the apparent terminal elimination half-life is approximately 6.5 hours.
When renal function is impaired, renal clearance is decreased in proportion to that of creatinine and thus the elimination half-life is prolonged, leading to increased levels of metformin in plasma.

Characteristics in specific groups of patients
<u>Renal impairment</u>
The available data in subjects with moderate renal insufficiency are scarce and no reliable estimation of the systemic exposure to metformin in this subgroup as compared to subjects with normal renal function could be made. Therefore, the dose adaptation should be made upon clinical efficacy/tolerability considerations.

DOSAGE AND ADMINISTRATION
Adults with normal renal function (GFR≥ 90 mL/min)
Reduction in the risk or delay of the onset of type 2 diabetes
• Metformin should only be considered where intensive lifestyle modifications for 3 to 6 months have not resulted in adequate glycaemic control.
• The therapy should be initiated with one tablet metformin hydrochloride extended release 500 mg once daily with the evening meal.
• After 10 to 15 days dose adjustment on the basis of blood glucose measurements is recommended (OGTT and/or FPG and/or HbA1c values to be within the normal range). A slow increase of dose may improve gastro-intestinal tolerability. The maximum recommended dose of Glucoxit XR 750 mg is 2 tablets (1500 mg) once daily with the evening meal.
• It is recommended to regularly monitor (every 3-6 months) the glycaemic status (OGTT and/or FPG and/or HbA1c value) as well as the risk factors to evaluate whether treatment needs to be continued, modified or discontinued.
• A decision to re-evaluate therapy is also required if the patient subsequently implements improvements to diet and/or exercise, or if changes to the medical condition will allow increased lifestyle interventions to be possible.

Monotherapy and combination with other oral antidiabetic agents:
• The usual starting dose of Glucoxit XR 750mg is one tablet daily given with the evening meal.
• After 10 to 15 days the dose should be adjusted on the basis of blood glucose measurements. A slow increase of dose may improve gastro-intestinal tolerability. The maximum recommended dose is 2 tablets once daily with the evening meal.
• In patients already treated with metformin tablets, the starting dose of Glucoxit XR should be equivalent to the daily dose of metformin immediate release tablets.
• If transfer from another oral antidiabetic agent is intended: discontinue the other agent and initiate Glucoxit XR at the dose indicated above.

Combination with insulin:
Metformin and insulin may be used in combination therapy to achieve better blood glucose control. The usual starting dose of Glucoxit XR is one tablet once daily up to a maximum of 1500 mg with the evening meal, while insulin dosage is adjusted on the basis of blood glucose measurements.

Elderly:
Due to the potential for decreased renal function in elderly subjects, the metformin dosage should be adjusted based on renal function. Regular assessment of renal function is necessary. Benefit in the reduction of risk or delay of the onset of type 2 diabetes mellitus has not been established in patients 75 years and older and metformin initiation is therefore not recommended in these patients.

Renal impairment
A GFR should be assessed before initiation of treatment with metformin containing products and at least annually thereafter. In patients at an increased risk of further progression of renal impairment and in the elderly, renal function should be assessed more frequently, e.g., every 3-6 months.

GFR (ml/min)	**Total maximum daily dose** (to be divided into 2-3 daily doses)*	**Additional considerations**
60-89	3000 mg	Dose reduction may be considered in relation to declining renal function
45-59	2000 mg	Factors that may increase the risk of lactic acidosis should be reviewed before considering initiation of metformin.The starting dose is at most half of the maximum dose.
30-44	1000 mg	
<30	-	Metformin is contraindicated.

Paediatric population
In the absence of available data, Glucoxit XR should not be used in children.

INTERACTION WITH OTHER MEDICINAL PRODUCTS AND OTHER FORMS OF INTERACTION
Concomitant use not recommended

<u>Alcohol</u>
Alcohol intoxication is associated with an increased risk of lactic acidosis, particularly in case of fasting, malnutrition or hepatic impairment.

<u>Iodinated contrast agents</u>
Metformin must be discontinued prior to or at the time of the imaging procedure and not restarted until at least 48 hours after, provided that renal function has been re-evaluated and found to be stable.

<u>Combination requiring precautions for use</u>
Some medicinal products can adversely affect renal function which may increase the risk of lactic acidosis, e.g. NSAIDs, including selective cyclo-oxygenase (COX) II inhibitors, ACE inhibitors, angiotensin II receptor antagonists and diuretics, especially loop diuretics. When starting or using such products in combination with metformin, close monitoring of renal function is necessary.

Medicinal products with intrinsic hyperglycaemic activity (e.g. glucocorticoids (systemic and local routes) and sympathomimetics).

More frequent blood glucose monitoring may be required, especially at the beginning of treatment. If necessary, adjust the metformin dosage during therapy with the other drug and upon its discontinuation.

Organic cation transporters (OCT)

Metformin is a substrate of both transporters OCT1 and OCT2.
Co-administration of metformin with:
• Inhibitors of OCT1 (such as verapamil) may reduce efficacy.
• Inducers of OCT1 (such as rifampicin) may increase gastrointestinal absorption and efficacy of metformin
• Inhibitors of OCT2 (such as cimetidine, dolutegravir, ranolazine, trimethoprime, vandetanib, isavuconazole) may decrease the renal elimination of metformin and thus lead to an increase in metformin plasma concentration.
• Inhibitors of both OCT1 and OCT2 (such as crizotinib, olaparib) may alter efficacy and renal elimination of metformin.

Caution is therefore advised, especially in patients with renal impairment, when these drugs are co-administered with metformin, as metformin plasma concentration may increase. If needed, dose adjustment of metformin may be considered as OCT inhibitors/inducers may alter the efficacy of metformin.

PREGNANCY
Pregnancy and Lactation
Uncontrolled diabetes during pregnancy (gestational or permanent) is associated with increased risk of congenital abnormalities and perinatal mortality. When the patient plans to become pregnant and during pregnancy, it is recommended that impaired glycaemic control or diabetes are not treated with metformin. For diabetes it is recommended that insulin should be used to maintain blood glucose levels as close to normal as possible to reduce the risk of malformations of the foetus.

Lactation
Metformin is excreted into human breast milk. No adverse effects were observed in breastfed new-borns/infants. However, as only limited data are available, breastfeeding is not recommended during

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GLUCOXIT XR TABLET

metformin treatment. A decision on whether to discontinue breast-feeding should be made, considering the benefit of breast-feeding and the potential risk to adverse effect on the child.

Fertility
Fertility of male or female rats was unaffected by metformin when administered at doses as high as 600 mg/kg/day, which is approximately three times the maximum recommended human daily dose based on body surface area comparisons.

Effects on ability to drive and use machines
Metformin monotherapy does not cause hypoglycaemia and therefore has no effect on the ability to drive or to use machines. However, patients should be alerted to the risk of hypoglycaemia when metformin is used in combination with other antidiabetic agents (e.g. sulphonylureas, insulin, or meglinitides).

Undesirable effects
In post marketing data and in controlled clinical studies, adverse event reporting in patients treated with Metformin XR was similar in nature and severity to that reported in patients treated with Metformin XR immediate release.

During treatment initiation, the most common adverse reactions are nausea, vomiting, diarrhoea, abdominal pain and loss of appetite, which resolve spontaneously in most cases.

The following adverse reactions may occur with GLUCOXIT XR.
Frequencies are defined as follow: very common:>1/10: common ≥ 1/100,<1/10:uncommon ≥ 1/1,000,<1/100:rare ≥1/10,000,<1/1,000: very rare <1/10,000.
Within each frequency grouping, adverse reactions are presented in order of decreasing seriousness.

Metabolism and nutrition disorders
Very rare:
- Lactic acidosis.
- Decrease of vitamin B12 absorption with decrease of serum levels during long-term use of metformin. Consideration of such an aetiology is recommended if a patient presents with megaloblastic anaemia.

Common: Vitamin B12 decrease/deficiency

Nervous system disorders
Common: Taste disturbance

Gastrointestinal disorders
Very common: Gastrointestinal disorders such as nausea, vomiting, diarrhoea, abdominal pain and loss of appetite. These undesirable effects occur most frequently during initiation of therapy and resolve spontaneously in most cases. A slow increase of the dose may also improve gastrointestinal tolerability.

Hepatobiliary disorders
Very rare: Isolated reports of liver function tests abnormalities or hepatitis resolving upon metformin discontinuation.

Skin and subcutaneous tissue disorders
Very rare: Skin reactions such as erythema, pruritus, urticaria.

CONTRAINDICATIONS
- Hypersensitivity to metformin or to any of the excipients listed in List of excipients.
- Any type of acute metabolic acidosis (such as lactic acidosis, diabetic ketoacidosis).
- Diabetic pre-coma.
- Severely reduced kidney function (GFR <30 mL/min).
- Acute conditions with the potential to alter renal function such as:
  - dehydration,
  - severe infection,
  - shock,
- Disease which may cause tissue hypoxia (especially acute disease, or worsening of chronic disease) such as:
  - decompensated heart failure,
  - respiratory failure,
  - recent myocardial infarction,
  - shock
- Hepatic insufficiency, acute alcohol intoxication, alcoholism

WARNING AND SPECIAL PRECAUTIONS

Lactic acidosis
Lactic acidosis, a very rare, but serious metabolic complication, most often occurs at acute worsening of renal function or cardiorespiratory illness or sepsis. Metformin accumulation occurs at acute worsening of renal function and increases the risk of lactic acidosis. In case of dehydration (severe diarrhoea or vomiting, fever or reduced fluid intake), metformin should be temporarily discontinued and contact with a health care professional is recommended.

Medicinal products that can acutely impair renal function (such as antihypertensives, diuretics and NSAIDs) should be initiated with caution in metformin-treated patients. Other risk factors for lactic acidosis are excessive alcohol intake, hepatic insufficiency, inadequately controlled diabetes, ketosis, prolonged fasting and any conditions associated with hypoxia, as well as concomitant use of medicinal products that may cause lactic acidosis.

Patients and/or care-givers should be informed of the risk of lactic acidosis. Lactic acidosis is characterised by acidotic dyspnoea, abdominal pain, muscle cramps, asthenia and hypothermia followed by coma. In case of suspected symptoms, the patient should stop taking metformin and seek immediate medical attention. Diagnostic laboratory findings are decreased blood pH (< 7.35), increased plasma lactate levels (>5 mmol/L) and an increased anion gap and lactate/pyruvate ratio.

Renal function:
A GFR should be assessed before treatment initiation and regularly thereafter, (see section Dosage and Administration). Metformin is contraindicated in patients with GFR<30 mL/min and should be temporarily discontinued in the presence of conditions that alter renal function, (see Contraindications).

Cardiac function
Patients with heart failure are more at risk of hypoxia and renal insufficiency. In patients with stable chronic heart failure, metformin may be used with a regular monitoring of cardiac and renal function. For patients with acute and unstable heart failure, metformin is contraindicated.

Elderly:
Due to the limited therapeutic efficacy data in the reduction of risk or delay of type 2 diabetes in patients 75 years and older, metformin initiation is not recommended in these patients.

Administration of iodinated contrast agents:
Intravascular administration of iodinated contrast agents may lead to contrast induced nephropathy, resulting in metformin accumulation and an increased risk of lactic acidosis. Metformin should be discontinued prior to or at the time of the imaging procedure and not restarted until at least 48 hours after, provided that renal function has been re-evaluated and found to be stable.

Surgery:
Metformin must be discontinued at the time of surgery with general, spinal or epidural anaesthesia. Therapy may be restarted no earlier than 48 hours following surgery or resumption of oral nutrition and provided that renal function has been re-evaluated and found to be stable.

Other precautions:
All patients should continue their diet with a regular distribution of carbohydrate intake during the day. Overweight patients should continue their energy-restricted diet.

Metformin may reduce vitamin B12 serum levels. The risk of low vitamin B12 levels increases with increasing metformin dose, treatment duration, and/or in patients with risk factors known to cause vitamin B12 deficiency. In case of suspicion of vitamin B12 deficiency (such as anaemia or neuropathy), vitamin B12 serum levels should be monitored. Periodic vitamin B12 monitoring could be necessary in patients with risk factors for vitamin B12 deficiency.

Metformin therapy should be continued for as long as it is tolerated and not contra-indicated and appropriate corrective treatment for vitamin B12 deficiency provided in line with current clinical guidelines.

The usual laboratory tests for diabetes monitoring should be performed regularly. Metformin alone never causes hypoglycaemia, although caution is advised when it is used in combination with insulin or other oral antidiabetics (e.g. sulphonylureas or meglitinides). The tablet shells may be present in the faeces. Patients should be advised that this is normal.

OVERDOSE
Hypoglycaemia has not been seen with metformin doses of up to 85g, although lactic acidosis has occurred in such circumstances. High overdose or concomitant risks of metformin may lead to lactic acidosis. Lactic acidosis is a medical emergency and must be treated in hospital. The most effective method to remove lactate and metformin is haemodialysis.

ROUTE OF ADMINISTRATION
For oral use

PACK SIZE
Glucoxit XR Tablet 750mg: 6 x 10's, 10 x 10's

STORAGE
Store at room temperature. Protect from moisture, freezing and excessive heat. Store below 30°C.

SHELF LIFE
2 years

PRODUCT REGISTRATION HOLDER
Duopharma Manufacturing (Bangi) Sdn. Bhd.
Lot No. 2, 4, 6, 8 & 10, Jalan P/7, Section 13, Bangi Industrial Estate, 43650 Bandar Baru Bangi, Selangor, Malaysia.

MANUFACTURER
Duopharma Manufacturing (Bangi) Sdn. Bhd.
Lot No. 2 & 4, Jalan P/7, Section 13, Bangi Industrial Estate, 43650 Bandar Baru Bangi, Selangor, Malaysia.

DUOPHARMA

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